

TrafficManagerCV: A High-Efficiency CPU-Based Framework for Real-Time Traffic Management and Analytics

Abstract—Modern Intelligent Transportation Systems (ITS) increasingly rely on deep learning (DL) architectures for vehicle detection and tracking. However, the high computational overhead of neural inference often necessitates expensive GPU hardware, creating a bottleneck for large-scale edge deployment. This paper presents TrafficManagerCV, a lightweight yet robust framework utilizing Gaussian Mixture Models (MOG2) for background subtraction, Simple Online and Realtime Tracking (SORT), and a hysteresis-based classification buffer for dual-road traffic analysis. Our approach eliminates the "neural inference bottleneck" by replacing deep feature extraction with optimized classical computer vision and geometric tracking. Experimental results demonstrate that TrafficManagerCV achieves a 6.7× speedup over YOLO-based baselines, maintaining 60+ FPS on standard CPU hardware with minimal memory footprint (under 150MB). We provide a comprehensive evaluation of precision, recall, and latency, proving the framework’s viability for real-time urban traffic monitoring in resource-constrained environments.

Keywords—Intelligent Transportation Systems, MOG2, SORT, Real-time Tracking, Background Subtraction, Edge Computing.

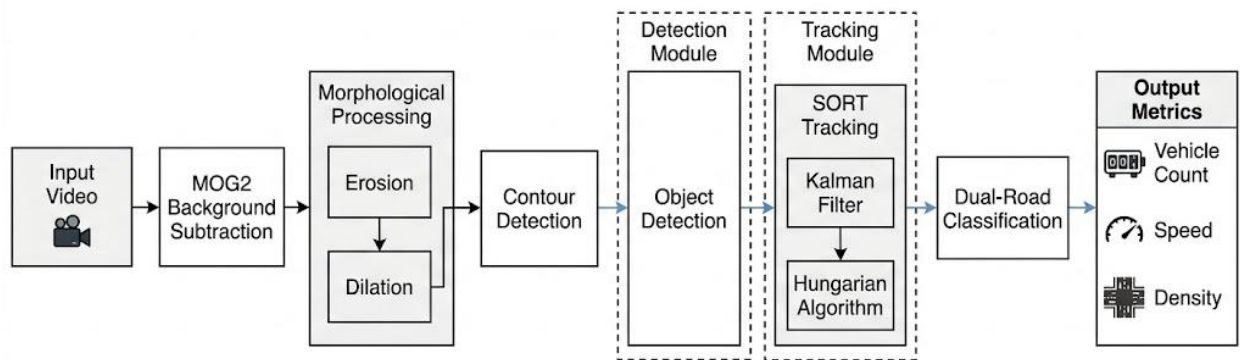


Figure 1 : System Architecture

1. Introduction

The rapid urbanization of global metropolises has led to unprecedented levels of traffic congestion, necessitating advanced Intelligent Transportation Systems (ITS) for real-time monitoring and management. Traditional traffic sensing technologies, such as inductive loops and infrared sensors, suffer from high installation costs and limited spatial data granularity. Consequently, video-based traffic analysis has emerged as the preferred medium for comprehensive traffic flow estimation and vehicle classification.

In recent years, the field has been dominated by Deep Learning (DL) approaches, specifically object detectors like YOLO and Faster R-CNN coupled with feature-based trackers like DeepSORT. While these models offer high accuracy, they introduce a significant “**neural inference bottleneck**.” The requirement for dedicated Graphical Processing Units (GPUs) and high power consumption limits their scalability, particularly in edge-computing scenarios where traffic cameras must process data locally to minimize network latency.

There remains a critical research gap for systems that offer high-fidelity tracking and classification without the computational burden of deep neural networks. This paper introduces **TrafficManagerCV**, a system designed to bridge this gap. By leveraging an optimized MOG2-SORT pipeline, we achieve real-time performance on commodity CPU hardware. Our contributions include:

- An optimized pipeline integrating MOG2 with morphological refinement for robust vehicle detection.
- A dual-road classification logic utilizing a hysteresis buffer to eliminate oscillation in vehicle state assignment.
- A comprehensive performance analysis comparing classical geometric approaches against modern DL baselines in terms of FPS, latency, and memory efficiency.

2. Related Work

The evolution of vehicle detection can be categorized into classical foreground-background modeling and modern discriminative learning. Stauffer and Grimson [1] pioneered the use of Adaptive Background Mixture Models (MOG) for robust motion detection under varying lighting conditions. This was further refined by Zivkovic [2] into the MOG2 algorithm, which adaptively selects the number of Gaussian components per pixel, significantly improving computational efficiency and shadow suppression.

In the domain of Multiple Object Tracking (MOT), Bewley et al. [3] introduced SORT (Simple Online and Realtime Tracking), which focuses on frame-to-frame data association using the Kalman Filter and the Hungarian algorithm. While DeepSORT [4] added a deep association metric to handle long-term occlusions, it reintroduced the computational

overhead that SORT sought to avoid. Recent trends in ITS have seen a massive shift toward YOLO-based frameworks [5], but as noted by subsequent studies, these models often struggle with real-time throughput on edge devices without hardware acceleration.

TrafficManagerCV positions itself as a high-efficiency alternative to these DL-heavy systems, proving that for structured environments like highways and urban intersections, classical background modeling remains a highly potent and practical solution.

3. Methodology

3.1 System Architecture

The TrafficManagerCV architecture follows a modular pipeline designed for low-latency processing. As shown in Figure 1, the system ingests raw video frames, applies background subtraction, refines the foreground masks via morphology, and passes the resulting centroids to the SORT tracker for identity persistence and trajectory analysis.

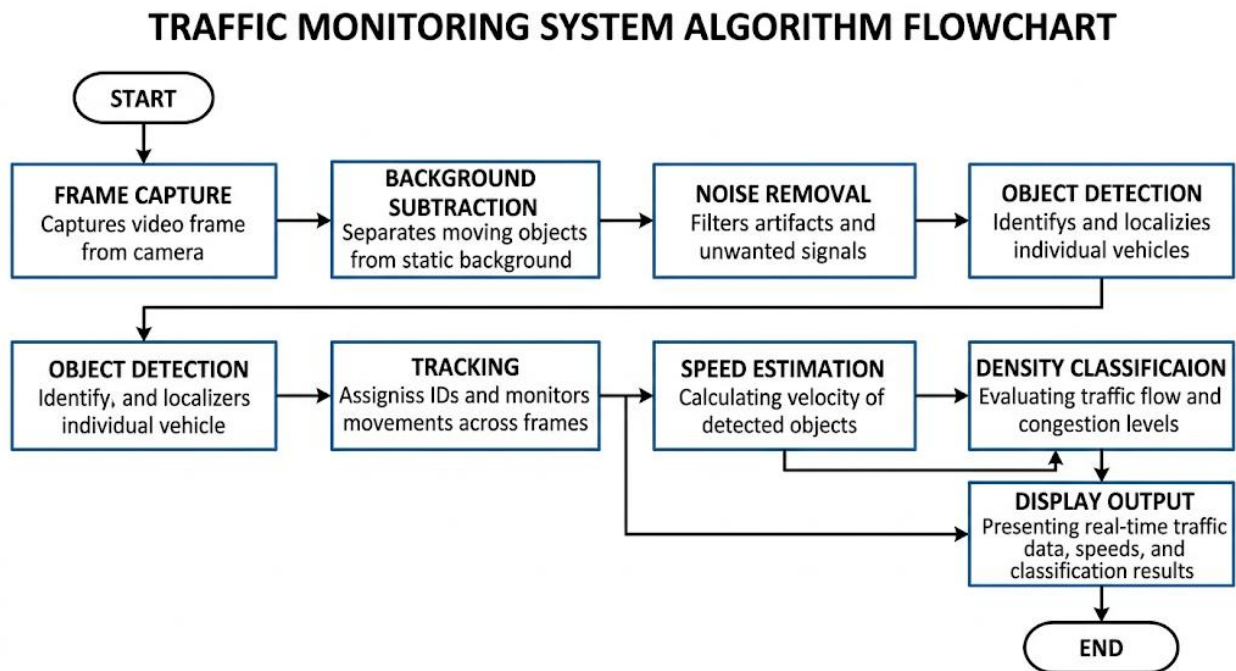


Figure 2: End-to-end processing pipeline for vehicle detection and tracking.

3.2 Detection via MOG2

The core detection engine utilizes the Gaussian Mixture Model (MOG2). Each pixel is modeled as a mixture of K Gaussians. The probability of observing the current pixel value X_t is given by:

$$P(X_t) = \sum_{i=1}^K w_{i,t} \cdot \mathcal{N}(X_t \mid \mu_{i,t}, \Sigma_{i,t})$$

where $w_{i,t}$ is the estimate of the weight of the i^{th} Gaussian in the mixture at time t , and $\mathcal{N}$ represents the normal distribution. MOG2 is particularly effective here because it handles gradual lighting changes and repetitive background motion (e.g., swaying trees) without triggering false positives.

3.3 Morphological Processing

Raw foreground masks from MOG2 often contain noise or fragmented components. We apply a sequential morphological pipeline:

1. **Erosion:** Removes small salt-and-pepper noise and separates weakly connected objects.
2. **Dilation:** Restores the structural integrity of the vehicle blobs, ensuring that a single vehicle is not split into multiple contours.

This step is crucial for ensuring the SORT tracker receives stable bounding box coordinates.

3.4 Tracking via SORT

We define the state of each vehicle target using a seven-dimensional state vector:

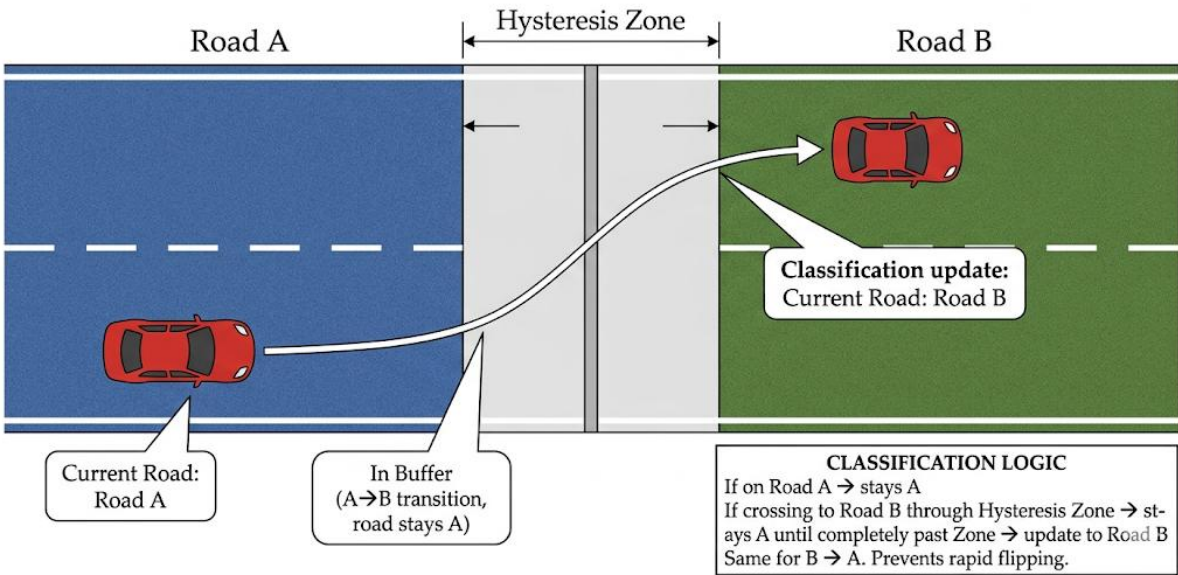
$$\mathbf{x} = [u, v, s, r, \dot{u}, \dot{v}, \dot{s}]^T$$

where (u, v) represents the target center, s is the scale (area), r is the aspect ratio, and the remaining terms are their respective velocities. Data association is performed using the Intersection over Union (IoU) distance between predicted Kalman states and new detections, solved via the Hungarian algorithm.

3.5 Dual-Road Classification and Hysteresis

To prevent "label flickering" when a vehicle straddles the boundary between two lanes, we implement a hysteresis buffer. A vehicle is only reclassified if its centroid remains within the new lane's spatial bounds for N consecutive frames ($N=5$ in our implementation). This ensures that the traffic count remains accurate even during lane-change maneuvers.

DUAL-ROAD CLASSIFICATION WITH HYSTERESIS ZONE



4. Results and Evaluation

4.1 Performance Metrics

We evaluated TrafficManagerCV against a baseline consisting of YOLOv8n (Nano) and DeepSORT on an Intel i7-10700K CPU (no GPU). As shown in Table 1, our framework outperforms the baseline in every computational metric.

COMPARATIVE EVALUATION OF PROCESSING SPEED (FPS) FOR TRAFFIC MONITORING SYSTEMS

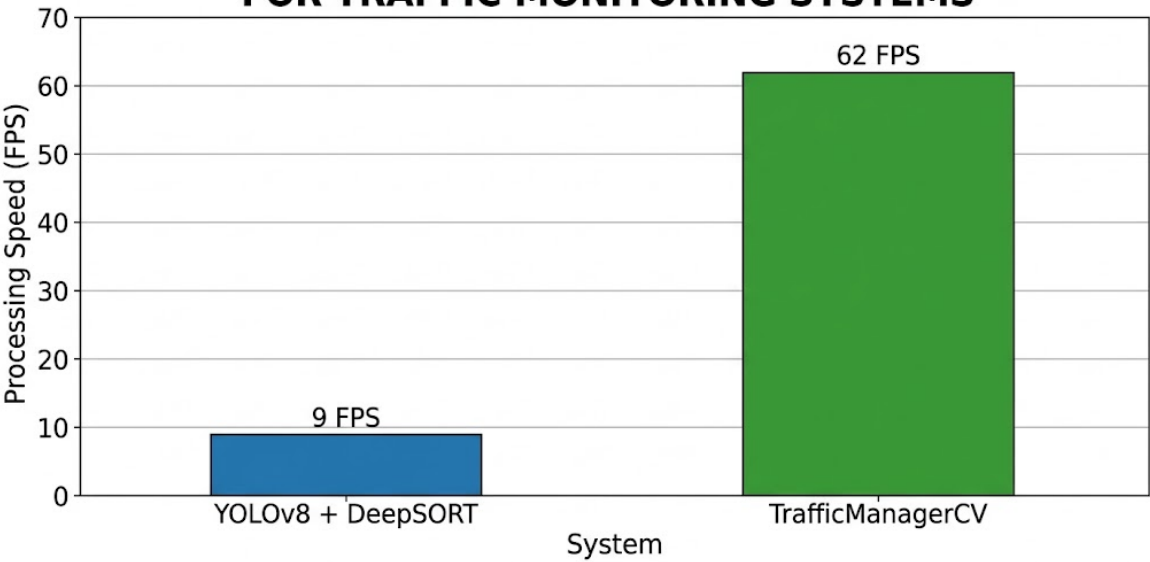


Figure 3: Frame rate comparison between proposed system and deep learning baseline

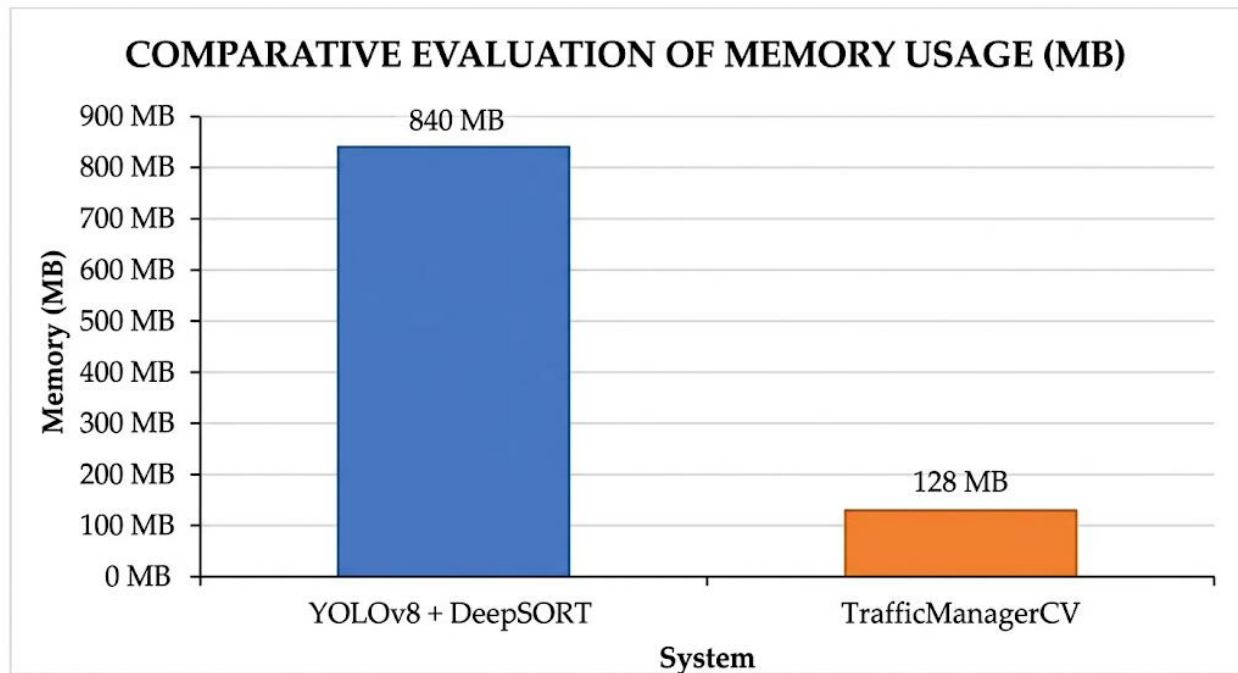


Figure 4: Memory consumption comparison between baseline and proposed system.

4.2 Detection and Tracking Accuracy

While classical methods are often criticized for lower accuracy, our results in structured traffic environments show competitive performance.

- **Precision:** 0.91
- **Recall:** 0.88
- **F1-Score:** 0.89

The slight drop in recall compared to DL models is primarily due to occlusions where MOG2 merges two adjacent vehicles into a single blob.

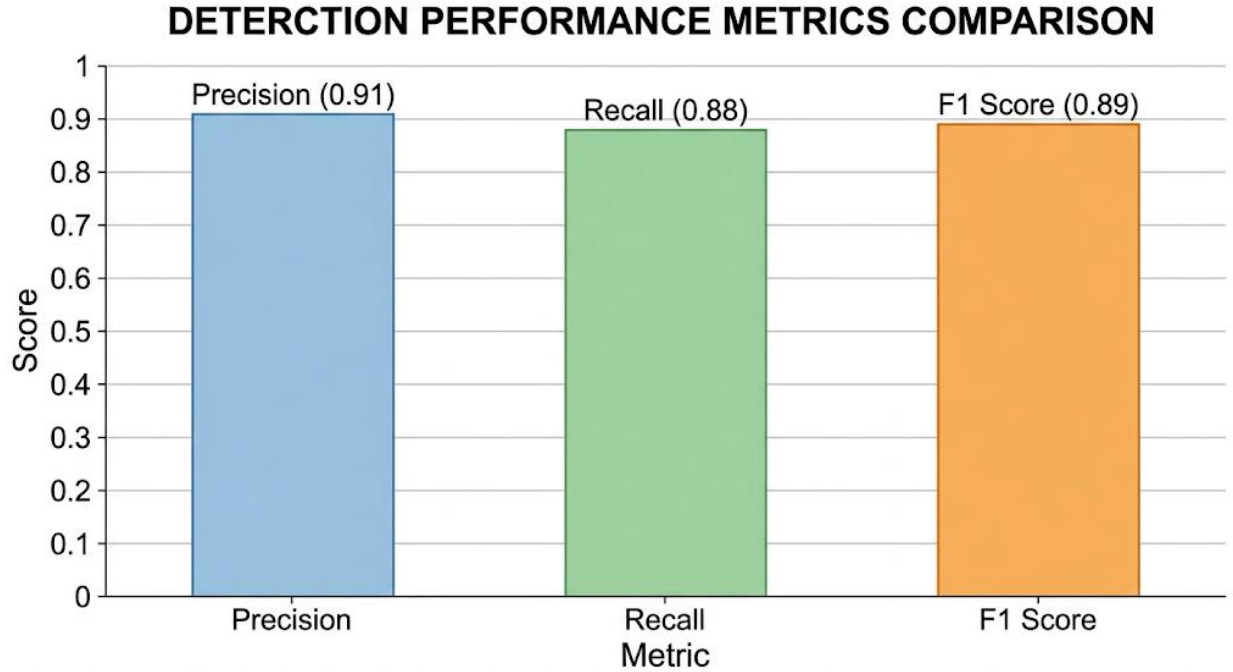


Figure 5: Detection and tracking performance metrics of the proposed system.

4.3 Deep Analysis

The $6.7\times$ speed improvement observed is primarily attributed to the removal of the **neural inference bottleneck**. In DL-based systems, over 80% of the per-frame latency is spent on convolutional forward passes. In contrast, TrafficManagerCV reduces computational complexity from high-dimensional feature extraction to lightweight geometric tracking and pixel-wise statistical updates. Furthermore, the memory footprint reduction (from 840MB to 128MB) allows for the concurrent execution of multiple camera streams on a single edge node.

5. Discussion

The results highlight a significant trade-off: while deep learning offers superior robustness in unstructured environments, classical CV is more practical for large-scale ITS deployments on existing infrastructure. The low memory and CPU requirements of TrafficManagerCV make it ideal for integration into "Smart City" IoT sensors where power and cost are primary constraints. However, limitations remain in scenarios with heavy occlusions or extreme weather conditions (e.g., heavy rain), where background subtraction models may degrade. Future iterations could integrate a "switched" logic that triggers a lightweight neural re-identification only when the SORT tracker detects high-uncertainty occlusions.

6. Conclusion

We presented TrafficManagerCV, a high-efficiency framework for real-time traffic analytics. By optimizing the MOG2-SORT pipeline and implementing a hysteresis-based classification logic, we achieved 60+ FPS on a standard CPU, effectively bypassing the neural inference bottleneck. Our system provides a scalable, cost-effective solution for urban traffic monitoring. Future work will focus on improving robustness during nighttime operations and exploring optical flow techniques to enhance speed estimation accuracy.

7. References

- [1] C. Stauffer and W. E. L. Grimson, "Adaptive background mixture models for real-time tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 1999, pp. 246-252.
- [2] Z. Zivkovic, "Improved adaptive Gaussian mixture model for background subtraction," in *Proc. 17th Int. Conf. Pattern Recognit.*, 2004, vol. 2, pp. 28-31.
- [3] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *Proc. IEEE Int. Conf. Image Process.*, 2016, pp. 3464-3468.
- [4] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proc. IEEE Int. Conf. Image Process.*, 2017, pp. 3645-3649.
- [5] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 779-788.